Alexander LeNail

alexander.lenail@wyss.harvard.edu | https://alexlenail.me | alexlenail | @alexlenail

Research

George Church Lab, Wyss Institute, Harvard Medical School
Postdoc *Sept. 2025 - present*

Developing next-generation AAV gene therapies for neurological diseases.

Myriam Heiman Lab, Picower Institute, MIT
MIT Computational & Systems Biology – PhD Student (PhRMA graduate research fellow) *Aug. 2019 - May 2025*

Thesis: Transcriptional Reprogramming to Reverse Aging-Associated Transcriptional Dysregulation in Neurons
Collected a single-cell multi-omics atlas of mouse brain aging. Analyzed it alongside public single cell atlases of brain aging and rejuvenation in mice, monkeys, and humans. Built ML models predicting chromatin accessibility and gene expression consequences of modulating TF levels. Used the model to predict TFs most likely to cause brain rejuvenation, and prepared 156 best TFs as an AAV pool. *In vivo* Perturb-Seq screen to validate those TFs' effects in the aged mouse motor cortex.

Google Brain
Research Intern *Summer 2019*

Predicting Transcription Factor Binding and Enhancer-Promoter Contact with Machine Learning on ENCODE/Roadmap data.

Ernest Fraenkel Lab, MIT Biological Engineering
MIT Biological Engineering – Research Associate *2016 - 2019*

Building computational infrastructure for AnswerALS & NeuroLINCS consortia. ML methods for the integration of 'omics data. Online data visualization / data portals for gene expression datasets.

Education

Massachusetts Institute of Technology *2019 - 2025*
Ph.D., Computational & Systems Biology *Cambridge, MA*

- Thesis Committee: Myriam Heiman, Jonathan Weissman, Manolis Kellis
- Coursework: Algorithms for Inference, Bayesian Modeling and Inference, Numerical Methods for Stochastic Modeling and Inference, Systems Biology, Chemical Biology, Protein Engineering, Topics in Computational & Systems Biology

Tufts University *2012 - 2016*
B.S., Computer Science *Somerville, MA*

- Concentrations: Machine Learning, Web Engineering, Computational Biology.

Professional Experience

2019 **Research Intern**, Google Brain
2016 - 2019 **Fraenkel Lab at MIT Biological Engineering**, Research Associate
2015 **Software Engineering Intern**, Benchling
2015 **Software Engineering Intern**, Google Search
2014 **KPCB Engineering Fellow**, Coursera
2013 **Software Engineering Intern**, Autodesk

Honors & Fellowships

2026 **ASGCT Career Development Award**, ASGCT
2026 **ALD Connect Emerging Investigator Award**, ALD Connect
2026 **Meritorious Abstract Travel Award**, ASGCT
2026 **Emerson Consequential Scholar**, Emerson Collective
2025 **2nd Best Poster**, GRC: Biology of Aging
2025 **Felicis Fellows in AI**, Felicis
2020 **Research Symbiont Award**, Wellcome Trust

Research Funding Awarded

2026 **Wyss Institute Validation Project**, TARGET-ALD: Targeted AAVs Restoring Glial Expression Therapeutically for ALD
2026 **Wyss Pollinator Fund**, Identifying AAV potency determinants to engineer better producer cells
2026 **AI Node Program Grant**, Foresight Institute
2026 **GENEWIZ Academic Grant, Gene Synthesis**, GENEWIZ
2025 **Wyss Director's Fund**, Hemoperfusion and Viral Filtration to Enhance Safety of Brain-Targeted Gene Therapy
2023 - 2025 **Graduate Research Fellowship**, PhRMA Foundation
2023 **GENEWIZ Week Grant**, GENEWIZ

Successful Grants Written

2024 **Glenn Foundation Discovery Award, American Federation for Aging Research (AFAR)**, Charting the Cellular Rejuvenation Landscape in Aging Neurons
2023 **Research Award**, Amaranth Foundation
2023 **Picower Institute Innovation Fund Engineering Award**, Picower Institute

PhD Publications (2019-2025)

Alexander LeNail, Raleigh Linville, Vanessa Farell, Saul Sauceda, Jaanak Prashar, Gaurav Arya, Preston Ge, Benjamin James, Jonathan Weissman, Manolis Kellis, Myriam Heiman. *Transcriptional Reprogramming to Reverse Aging-Associated Transcriptional Dysregulation in Neurons*. In Preparation.

Joseph M Replogle, Reuben A Saunders, Angela N Pogson, Jeffrey A Hussmann, **Alexander LeNail**, ..., Jonathan S Weissman. *Mapping information-rich genotype-phenotype landscapes with genome-scale Perturb-seq*. Cell, 2022.

Research Associate Publications (2016-2019)

Stanislav Tsitkov, Kelsey Valentine, Velina Kozareva, Aneesh Donde, Aaron Frank, Susan Lei, the **Answer ALS Consortium**, Jennifer E. Van Eyk, Steve Finkbeiner, Jeffrey D. Rothstein, Leslie M. Thompson, Dhruv Sareen, Clive N. Svendsen & Ernest Fraenkel. *Disease related changes in ATAC-seq of iPSC-derived motor neuron lines from ALS patients and controls*. Nature Communications, 2024.

Divya Ramamoorthy, Kristen Severson, Soumya Ghosh, Karen Sachs, **Answer ALS Consortium**, Jonathan D Glass, Christina N Fournier, Pooled Resource Open-Access ALS Clinical Trials Consortium, ALS/MND Natural History Consortium, Todd M Herrington, James D Berry, Kenney Ng, Ernest Fraenkel. *Identifying patterns in amyotrophic lateral sclerosis progression from sparse longitudinal data*. Nature Computational Science, 2022.

Answer ALS Consortium. *Answer ALS, a large-scale resource for sporadic and familial ALS combining clinical and multi-omics data from induced pluripotent cell lines*. Nature Neuroscience, 2022.

The NeuroLINCS Consortium. *An integrated multi-omic analysis of iPSC-derived motor neurons from C9ORF72 ALS patients*. iScience, 2022.

Alexander LeNail. *NN-SVG: Publication-Ready Neural Network Architecture Schematics*. Journal of Open Source Software, 2019.

Maxwell P. Gold, **Alexander LeNail**, Ernest Fraenkel. *Shallow Sparsely-Connected Autoencoders for Gene Set Projection*. Pacific Symposium of Biocomputing, 2019.

Tenley C Archer, Tobias Ehrenberger, Filip Mundt, Maxwell P Gold, Karsten Krug, Clarence K Mah, Elizabeth L Mahoney, Colin J Daniel, **Alexander LeNail**, Divya Ramamoorthy, Philipp Mertins, DR Mani, Hailei Zhang, Michael A Gillette, Karl Clauser, Michael Noble, Lauren C Tang, Jessica Pierre-François, Jacob Silterra, James Jensen, Pablo Tamayo, Andrey Korshunov, Stefan M Pfister, Marcel Kool, Paul A Northcott, Rosalie C Sears, Jonathan O Lipton, Steven A Carr, Jill P Mesirov, Scott L Pomeroy, Ernest Fraenkel. *Proteomics, Post-translational Modifications, and Integrative Analyses Reveal Molecular Heterogeneity within Medulloblastoma Subgroups*. Cancer Cell, 2018.

Aude Nicolas,...(many authors), **Alexander LeNail**, ...Bryan J. Traynor, John E. Landers. *Genome-wide Analyses Identify KIF5A as a Novel ALS Gene*. Neuron, 2018.

Alexandra B. Keenan, ...(many authors), **Alexander LeNail**, ... Ajay Pillai. *The Library of Integrated Network-Based Cellular Signatures NIH Program: System-Level Cataloging of Human Cells Response to Perturbations*. Cell Systems, 2018.

Alexander LeNail, Ludwig Schmidt, Jonathan Li, Tobias Ehrenberger, Karen Sachs, Stefanie Jegelka, Ernest Fraenkel. *Graph-Sparse Logistic Regression*. Neural Information Processing Systems: Discrete Structures in Machine Learning Workshop, 2017.

Murodzhon Akhmedov, **Alexander LeNail**, Francesco Bertoni, Ivo Kwee, Ernest Fraenkel & Roberto Montemanni. *A Fast Prize-Collecting Steiner Forest Algorithm for Functional Analyses in Biological Networks*. Integration of AI and OR Techniques in Constraint Programming, 2017.

Invited Talks & Posters

2026.05.12. *AAV Gene Therapy For Adrenoleukodystrophy with Transferrin-Receptor Dependent Capsids*. ASGCT 2026 Poster, Meritorious Abstract Travel Award

2025.11.07. *Gene Therapy for Adrenoleukodystrophy with New AAV Capsids*. ALD Connect Poster

2025.08.08. *Transcriptional Reprogramming to Reverse Aging-Associated Transcriptional Dysregulation in Neurons*. Biology of Aging GRS (Aug. 2-3): invited talk (Aug. 2) & poster. GRC (Aug. 3-8): poster & **2nd Best Poster award**

2025.04.30. *Computational Design of Transcription Factor Gene Therapies to Reverse Age-Associated Neurodegeneration*. Flash talk & poster: Schmidt Center Symposium on Biomedical Science and AI, Broad Institute

2024.11.18. *Charting the Cellular Rejuvenation Landscape in Aging Neurons*. Talk: MIT BCS departmental seminar series

2024.06.01. *Charting the Cellular Rejuvenation Landscape in Aging Neurons*. Invited talk: Gordon Research Seminars: Systems Aging

2024.05.09. *Optimizing AAV Dosing for In Vivo Screens in the Mouse Brain*. ASGCT 2024 Poster

2023.10.23. *Charting the Cellular Rejuvenation Landscape in Aging Neurons*. Poster: Cutting Edge Approaches to Studying the Aging Brain Symposium

2022.04.01. *ODE Parameter Inference*. Guest Lecture, Harvard CS 282r: Advanced Topics in Machine Learning

2017.12.08. *Graph-Sparse Logistic Regression*. Spotlight talk: NeurIPS workshop, Discrete Structures in Machine Learning

2016.03.31. *Machine Learning as Function Approximation*. Guest Lecture, Tufts CS 135: Machine Learning

Teaching Experience

Spr. 2020	**Introduction to Biology**, Teaching Assistant	*MIT*
Fall 2015	**Bioinformatics**, Teaching Assistant	*Tufts*

Mentoring

2026	**Nasser Al-Rais**, currently an undergraduate at Cambridge University
2021 - 2024	**Gaurav Arya**, now PhD student in CS at CMU
2022 - 2024	**Saul A. Vega Sauceda**, now Software Engineer at Flatiron Health
2023	**Jaanak Prashar**, now undergraduate at Stanford
2022 - 2023	**Stephanie P. Howe**, now MS student in Computational Biology at MIT
2022	**Jorge Lopez-Nava**, now post-bac Research Fellow at the Broad Institute
2022	**Andrew Zhao**, now MS student in Computational Biology at MIT

Outreach & Professional Development

Service and Outreach

2023 - 2025	**NeuroLunch**, Lead Organizer of main Neuroscience trainee talk series at MIT.
2023	**MIT CSB Application Assistance Program**, member

Professional Memberships

American Society for Gene & Cell Therapy (ASGCT)